



Contents lists available at ScienceDirect

Internet of Things

journal homepage: www.elsevier.com/locate/iot

A survey on internet of things security: Requirements, challenges, and solutions

Hamed HaddadPajouh^a, Ali Dehghantanha^a, Reza M. Parizi^{b,*},
Mohammed Aledhari^b, Hadis Karimipour^c

^aCyber Science Lab, School of Computer Science, University of Guelph, Ontario, Canada

^bCollege of Computing and Software Engineering, Kennesaw State University, GA, USA

^cSchool of Engineering, University of Guelph, Ontario, Canada

ARTICLE INFO

Article history:

Received 31 August 2019

Revised 5 November 2019

Accepted 5 November 2019

Available online xxx

Keywords:

Internet of things

Security requirements

IoT security

IoT architectural threats

Survey

ABSTRACT

Internet of Things (IoT) is one of the most promising technologies that aims to enhance humans' quality of life (QoL). IoT plays a significant role in several fields such as health-care, automotive industries, agriculture, education, and many cross-cutting business applications. Addressing and analyzing IoT security issues is crucial because the working mechanisms of IoT applications vary due to the heterogeneity nature of IoT environments. Therefore, discussing the IoT security concerns in addition to available and potential solutions would assist developers and enterprises to find appropriate and timely solutions to tackle specific threats, providing the best possible IoT-based services. This paper provides a comprehensive study on IoT security issues, limitations, requirements, and current and potential solutions. The paper builds upon a taxonomy that taps into the three-layer IoT architecture as a reference to identify security properties and requirements for each layer. The main contribution of this survey is classifying the potential IoT security threat and challenges by an architectural view. From there, IoT security challenges and solutions are further grouped by the layered architecture for readers to get a better understanding on how to address and adopt best practices to avoid the current IoT security threats on each layer.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

Internet of Things (IoT) paradigm has brought the attention of massive service providers, businesses, and industries such as Healthcare, Autonomous Vehicles, Smart Grids, Digital Agriculture, and many others. IoT enables objects to hear, listen, talk, and act in smart manners for which IoT usage is expected to be more pervasive by the end of 2020 as a part of Industry 4.0 revolution [1].

"Things" refer to a set of objects that could communicate and act with/without human interaction for which the IoT can be seen as robotic of things (RoT). The things can be sensors, automobiles, refrigerators, washer machines, and many other [2]. Understanding the IoT space requires defining the IoT layers and elements to describe the possible IoT architectures based on the required services and fields. There have been different architectures proposed for IoT environments. Generally,

* Corresponding author.

E-mail addresses: hhaddadp@uoguelph.ca (H. HaddadPajouh), adehghan@uoguelph.ca (A. Dehghantanha), rparizi1@kennesaw.edu (R. M. Parizi), maledhar@kennesaw.edu (M. Aledhari), hkarimi@uoguelph.ca (H. Karimipour).

<https://doi.org/10.1016/j.iot.2019.100129>

2542-6605/© 2019 Elsevier B.V. All rights reserved.

Table 1
Abbreviation glossary.

Abbreviation	Defination
API	Application Programming Interface
GPS	Global Positioning System
NFC	Near-Field Communication
DoS	Denial-of-Service
RFID	Radio-frequency identification
GPRS	General Packet Radio Services
SSL	Secure Sockets Layer
WSN	Wireless Sensor Network
SCM	Supply Chain Management
SOA	Service Oriented Application
EPC	Electronic Product Code
uID	Unified Information Devices
SCMaaS	Supply Chain Management as a Service
CPS	Cyber Physical System
REST	Representational State Transfer
SOAP	Simple Object Access Protocol
XSS	Cross-Site Scripting
DFA	Deterministic finite automaton
CoAP	Constrained Application Protocol
DTLS	Datagram Transport Layer Security
NDP	Neighbor Discovery Protocol
RPL	Routing Protocol for Low-Power and Lossy Networks
SSO	Single Sign On
IETF	Internet Engineering Task Force
DNSSEC	Domain Name System Security
DoT	DNS over TSL

such architectures are grouped into three classes as follows (a glossary for each abbreviation and acronym that mentioned in this paper are shown in Table 1):

1. Three-layered architecture: This class is the most common type among most of the publications and fellow researchers [3]. It is structured through top-down approach layers named as (a) Application layer in which both application and services functions operate there (b) Network/ Transmission layer (c) Perception/Edge layer which is related to the things or end-point devices of the architecture.
2. Four-layered architecture: The structure of this class is very similar to the three-layered architecture, while application and service are defined with a separate layer in the architecture.
3. Five-layered architecture: In this architecture two additions are considered (a) a business layer above application layer for delivering a wide range of functionalities in each different domain of IoT, or (b) a sub-layer in the edge layer for those applications of IoT which may be very critical at the device level.

1.1. Application layer

Although there is no comprehensive standard is presented for the IoT application layer, this layer can provide a range of services in different applications. For example, application of IoT in smart cities and home [4,5], smart grids [6–8], healthcare [9,10], autonomous vehicles [11–13]. The application layer can also operate as middleware [14], a communication protocol, and cloud computing for service support; thus, security concern would be different based on the environment and industry of the application. As can be seen in Fig. 1 in the application layer architecture different components are defined, and each component operations depend on the application of the environment. For instance, in healthcare for medical records retrieval, there will be a need for special application programming interface (API), or special applications as binary in client and server sides [15]. Most application security architectures focus on securing the CoAP protocol using DTLS. Whereas, the other application security architectures have proposed their model based on the encryption of HTTP payloads [16–20].

1.2. Network layer

In the network layer data transmission among the other layers is managed. This layer also provides access to the perception layer via different standards and protocols like IEEE 802.x, Global Positioning System (GPS), and Near-Field Communication (NFC). As shown in Fig. 2, this layer is also backed by a cloud back-end infrastructure, mobile devices, and the Internet protocol [21]. Moreover, the network layer can be treated with different aspects based on the applied environment. However, the most common security mechanism in the network layer of IoT architectures includes Block-chain technology, Intelligence Intrusion Detection Systems, and key management and encryption systems [22–27] (Fig. 3).

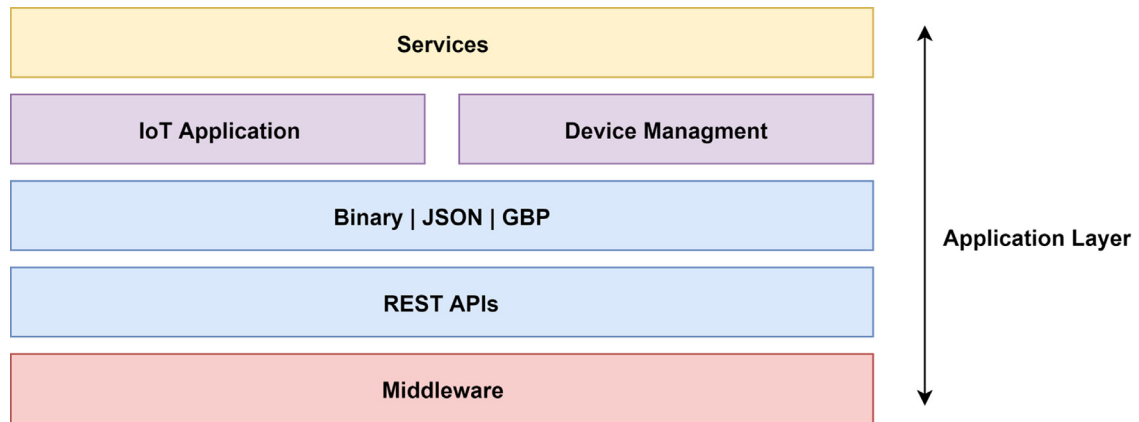


Fig. 1. Application layer components in IoT Architecture.

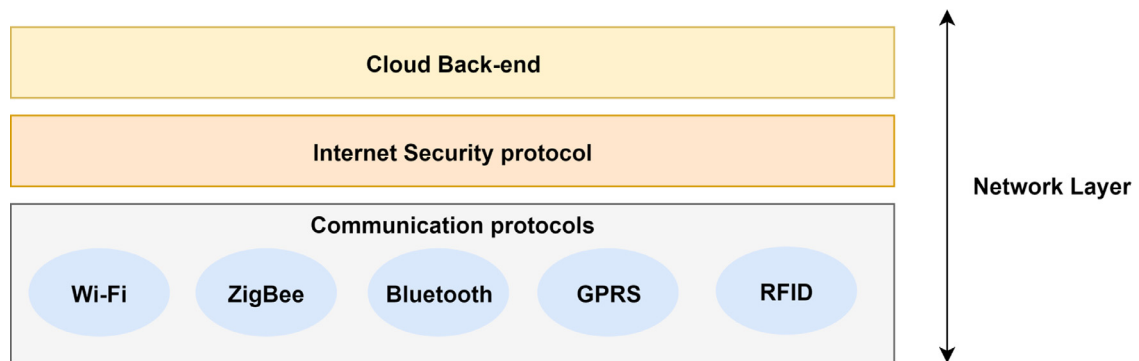


Fig. 2. Network layer components in IoT Architecture.

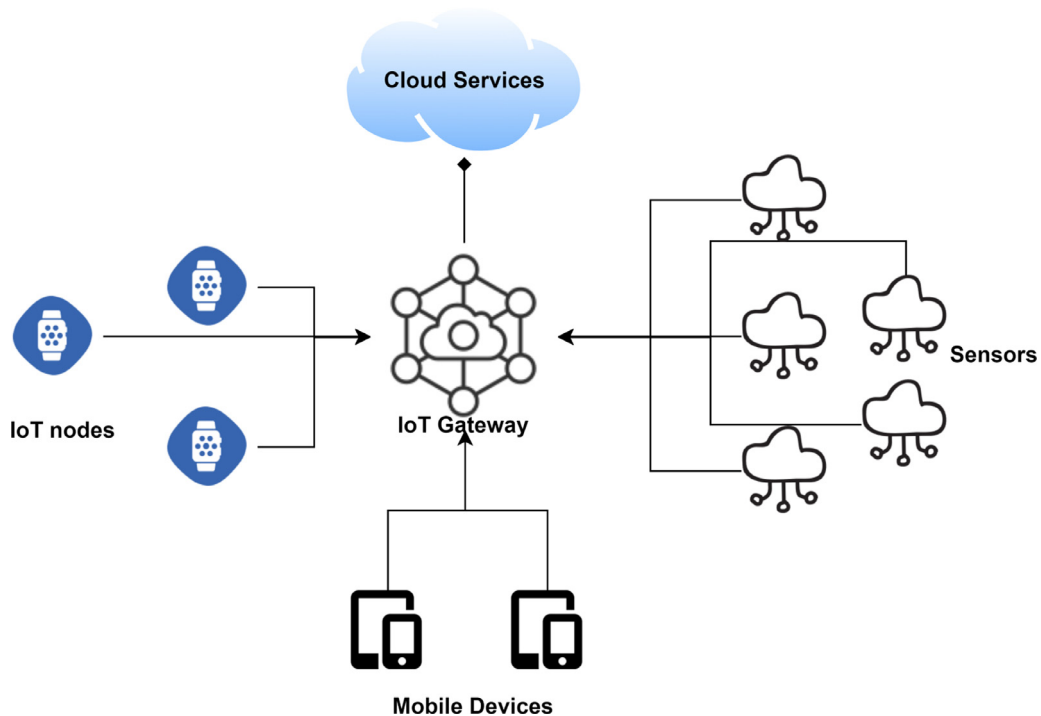


Fig. 3. Edge (device) layer components in IoT Architecture.

Table 2

Recent work on security vulnerabilities in the IoT architecture.

Layer	Vulnerabilities	Environment	References
Edge-Layer	Deficient physical security, Insufficient energy harvesting	Big Data Processing, m-Health, Military, Physical Security	[34–39]
Network Layer	Inadequate authentication, Improper encryption, Unnecessary open ports	Financial, Smart-city, Industrial, Physical Security, Cloud Computing	[40,41],[42–45]
Application Layer	Improper patch management capabilities, Weak programming practices (e.g. root user, lack of SSL, plain text password, backdoor, ect.), Insufficient audit mechanism	Military, Fog Computing, Smart-city, Physical Security	[34,42,46–50]

1.3. Edge layer

In this layer, end-users (cloud-edge) IoT devices can interact with the client or their working domains like sensors, smart meters or IoT edge layer servers of a gateway which has a coordinator role in the working domain. Due to the physical exposure of edge-layer in the IoT architecture, this layer faces a large number of attacks. The most common security components that applied in this layer include Multi-factor authentication mechanism, end-point anti-malware solution, secure channeling, and machine learning-based solutions for anomaly detection in the cloud-edge devices [28–33].

The disruptive usage of IoT devices with increasing computational resources in different application domains has led to a wide range of vulnerabilities in IoT environments. These vulnerabilities can cause fatal failures and information lost in different domains. Thus, the security of IoT environments has been considered as one of the hot topics in recent years, drawing research community's attention. As evidence of this trend, Table 2 shows the recent work on security vulnerabilities in the IoT architecture.

From the security aspect, regardless of the architecture of the IoT environments, there are a wide range of threats associated with each layer of IoT architecture. Each threat needs an appropriate security solution to prevent, detect and compensate the attack. Hence, from the top-down approach, security threats in the application layer usually include information stealing, malware propagation and botnet attack on web services. The network layer is very favorable to attackers of IoT environments to exploit vulnerabilities, which exist in protocols and standards. Denial of Service (DoS), session hijacking, and corruption are the most common threats occur in protocols and standards [51]. In cloud computing components of IoT, users encounter a wide range of security threats and vulnerabilities such as misconfiguration, lack of good identity management, inappropriate data access control, infrastructure security, and data privacy violation [52]. In the edge-layer, due to the predictable location of the connected devices in IoT environment and having computation resources limitation, this layer faces a whole range of different threats including node sabotaging, node failure, node disconnection, offline information gathering, false node message corruption, exhaustion, Sybil, jamming, tampering, and collisions [36].

Up to this point, there are several high-quality survey papers published in the literature in the domain of IoT covering security and privacy, architecture and applications [1,3,51–54]. In this paper, we attempt to cover significant works, which conducted in the last 10 years toward identifying security requirements, challenges and include those that give appropriate solutions to mitigate IoT security problems. In addition, we propose a three layer-based taxonomy for mapping security requirements in each functional level of IoT architecture for the readers with a diverse background in IoT. Moreover, we perform a descriptive analysis of the identified security requirements, challenges, and solutions from a top-down approach upon the essential security needs in different applications of the IoT. Most of the highly referred research papers consider the three-layer architecture as an essential structure for IoT [55–58], which is well inline with the objectives of this research.

The specific contributions of this paper include:

- Identifying and analyzing security requirements and vulnerabilities of the IoT environments, as well as a taxonomy based on a three layer architecture for IoT security requirements and vulnerabilities.
- Presenting and evaluating security challenges and their corresponding solutions of the IoT environments, as well as a taxonomy based on a three layer architecture for security challenges and solutions.

The rest of this paper is structured as follows. In Section 2, we deal with the IoT applications and security requirements. In Section 3, we present IoT security and privacy challenges. Section 4 provides security solutions for each challenge in a three-layer architectural way, and finally in Section 5 we discuss IoT security and privacy challenges and requirements solutions in the current and future applications based on the identified requirements.

2. IoT applications and security requirements

To mitigate the cyber threat impacts in different IoT applications, security requirements need to properly provided. In this view, this section first proposes a taxonomy of IoT applications based on the degree of involvement in each layer. Afterward, it gives the essential security requirements from layered-based architecture.

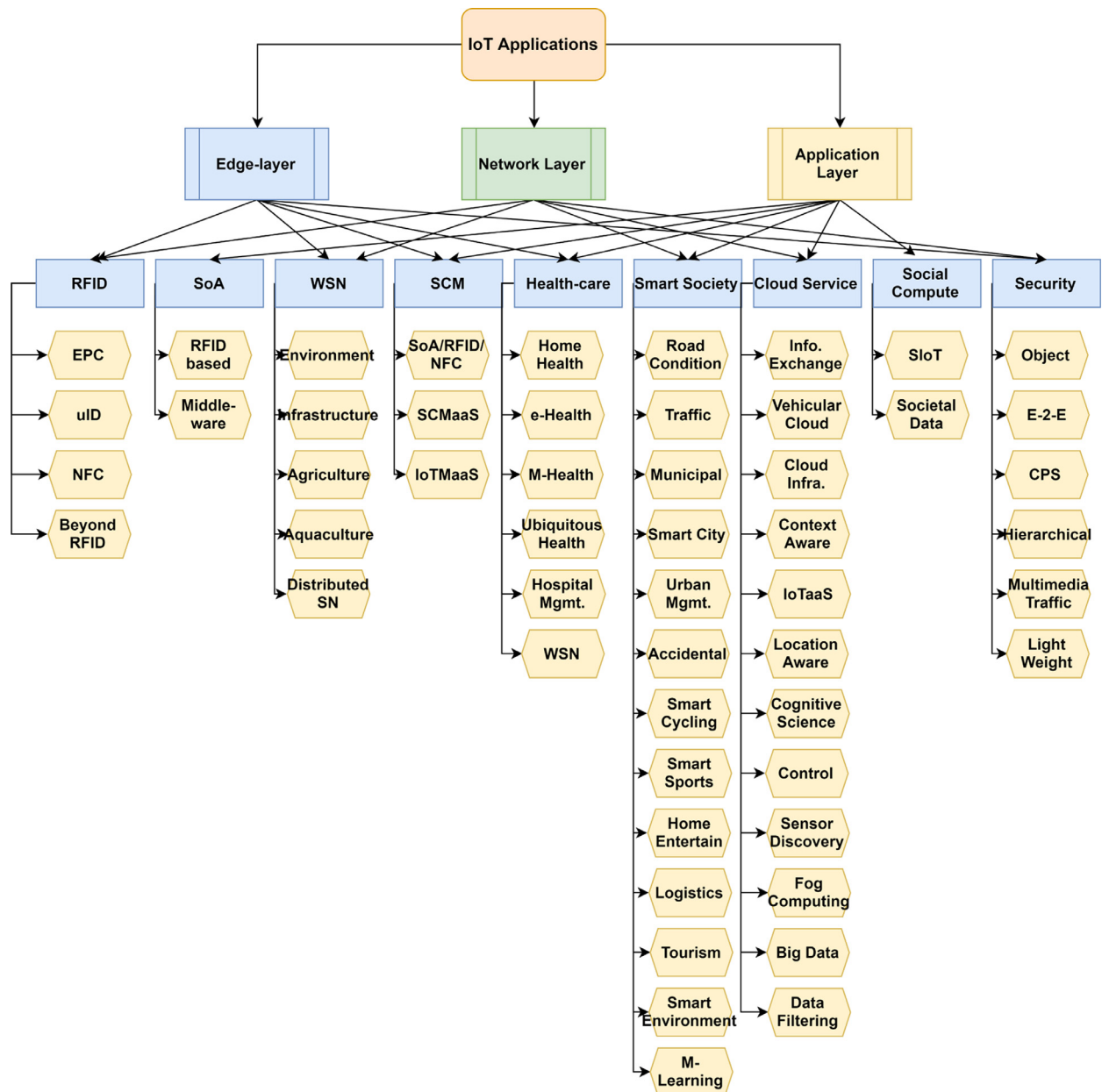


Fig. 4. IoT Applications Taxonomy based on domain applications [3].

2.1. IoT applications

Since each IoT component is able to operate without any third-party intervention, its role can be determined in every semi-automated/fully-automated environment. Healthcare, transportation, and automotive industries were the early adopter's domains applied IoT with the considerable positive impacts. However, we adopted [3] classification to provide a three-layered based taxonomy for the IoT applications. As it can be seen in Fig. 4, most of IoT applications divided into nine classes namely: Radio Frequency Identification (RFID) oriented application, Service-oriented Application (SoA), Wireless Sensor Network (WSN), Supply Chain Management (SCM), Healthcare include (m-health and e-health), Smart Society, Cloud-based Services, Social compute and Security. Each of these classes is specifically divided into the real domain of applications which involved with one or more layers of IoT architecture. These domain applications described as follow:

2.1.1. RFID

Connecting the objects by Radio Frequency Identification technology is well-known technology from the early 2000s, researchers now do believe that this architecture is related to IoT environment [4]. The RFID technology is highly relied on

the edge layer for environmental contacts and the network layer for handling the communications. To have safe and secure functionalities, this two-layer security mechanism should be validated.

2.1.2. Service Oriented Application

SOA is a flexible working architectural approach applied for business processes automation by integrating different components in each environment over the enterprise network. However, with increasing IoT service applications, the environment may involve a wide range of third-party components, which require the IoT service architecture to have a robust integration capability and interoperability in a heterogeneous environment, along with the features that can quickly address business needs through a rapid development or the reuse/integration of existing resources [59].

2.1.3. Wireless Sensor Network

Wireless Sensor Network (WSN) which counts as the IoT ancestor, is one of the prominent components of IoT environment [60–62]. This component includes a set of sensor nodes (mote) managed by a coordinator (sink) by employing a wide range of protocols organization in multi-layer structure [63].

2.1.4. Supply Chain Management

IoT plays a vital role in the supply chain management process in most application domains. This process includes the flow of goods from suppliers to the consumers in an automated fashion [64,65].

2.1.5. Healthcare

IoT application in the healthcare system has dramatically increased by recent developments. For example, the wearable devices technology is one of the most essential technologies used in IoT-based healthcare system. Since the wearable devices (sensors) are applied to collect sensitive data and may use in adversary environments, hence, needs for security mechanisms to avoid malicious interaction with the coordinator system is an essential requirement [21].

2.1.6. Smart society

The idea of smart society is to take accurate city information in the right situation at the right time in order to take the optimum decision regards to the event. A huge number of wired and wireless sensors are taken into consideration to establish an IoT-based Smart City. The core challenge to succeed in building smart societies is to link all the city data generated by various smart systems and sensors in one place [66]. In addition, security concern in terms of privacy in a vast amount of collected data for usage is considered a challenging issue.

2.1.7. Cloud-based services

IoT Cloud-based services which perform as back-end infrastructure of IoT are classified into the stationary and mobile application. The nature of resource limitation and short coverage in communication and mobility of IoT devices, resulting in a high demand for packet forwarding in cloud-based services in IoT environments.

2.1.8. Social compute

Social IoT (SIoT) is similar to a social network structure where each node mapped to a connected device with an ability to establish social relationships with other things (connected devices) in a fully-automated way under pre-defined rules that monitor by a coordinator [67].

2.1.9. Security

IoT devices play a vital role in maintaining and monitoring objects and environments. Monitoring environments by IoT devices need a reliable security mechanism for preventing data breach and maintaining data confidentiality on the devices. In addition, for real-time monitoring these devices need to run a light-weight cryptographic algorithm due to lower overhead.

2.2. IoT security requirements

IoT environments and related properties cause several security issues that have raised IoT users' concerns. The common properties in most of the IoT environments can be defined as follows:

Mobility: Almost all IoT devices are applied in a mobile way and are connected to the Internet via a wide range of providers.

Wireless communication: IoT devices generally connect to the Internet via a myriad of wireless links, including Bluetooth, 802.11, WiMAX, Zigbee, etc. Exposing wireless signal in public-eye frequency has been a security concern in recent years.

Embedded use: Major IoT devices have a single functionality. Accordingly, the detection of patterns in data transmission is often presented in a unique mode and can be considered as a potential vulnerability.

Diversity of component: There is a wide range of inconsistencies in terms of connected devices type and topology in the IoT. Privacy designs must accommodate even the simplest of devices.

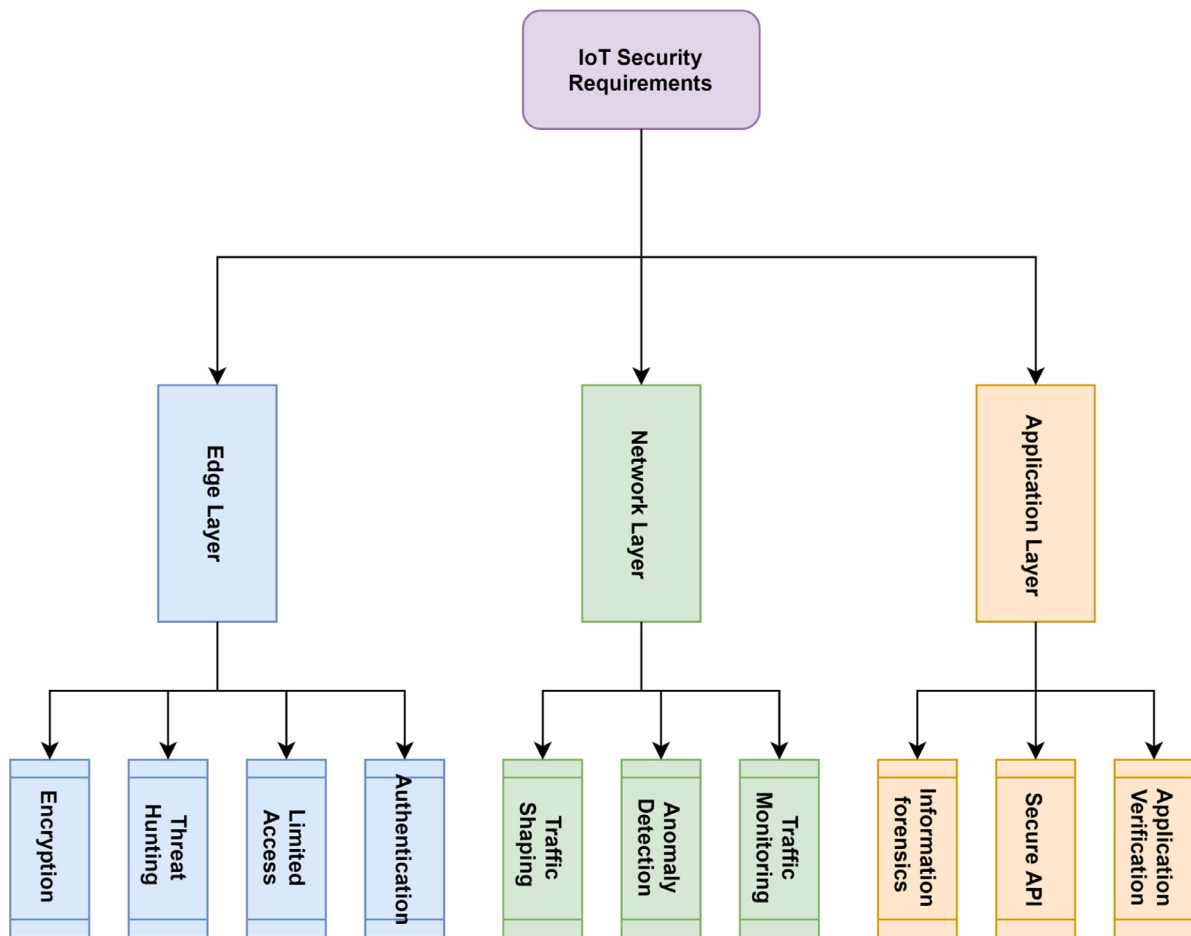


Fig. 5. IoT security requirements taxonomy based on layered architecture .

Table 3

IoT security vulnerabilities and requirements relationship.

Vulnerability	Security requirement	Architecture layer
Deficient Physical Security	Limited Access	Edge-layer
Insufficient energy harvesting	Threat Hunting	Edge-layer
Inadequate authentication	MAF Authentication	Edge-layer
Improper encryption	Traffic Monitoring, Encrypt the payload	Network-Layer
Unnecessary open ports	Traffic Monitoring, Anomaly Detection, Traffic Shaping	Network-Layer
Weak programming practices	Secure API	Application Layer
Insufficient audit mechanism	Application Verification	Application Layer
	Application Layer	
Improper patch management	Information forensics	Application Layer

Scalability: The number of connected devices is growing dramatically day by day and IoT clients can not monitor the privacy of own data under computation of these devices.

According to the above properties, we classified the security requirements in the IoT environment for each layer separately. As it can be seen in Fig. 5, the security requirements for the Application layer include: Application verification, Secure APIs, and Data Analytics. In the Network layer, security requirements are divided into three groups: Traffic Monitoring, Anomaly detection, and Traffic Shaping. The Edge layer's security requirements are classified as Authentication, Limited access, Threat hunting and Encryption.

All in all, we assigned each layer of the IoT architecture to the corresponding application based on each layer's service that it delivers, as follows (Table 3):

- Application-layer: Most IoT services in the application layer related to end-user interaction with delivery services. Therefore, we provided a taxonomy based on the application services defined for user interaction in the reviewed research

papers. For instance, most IoT users are a client of a cloud service for storing their data, thus, cloud service application is positioned for the application layer.

- Network-layer: Although almost every IoT component has network connectivity for data transmission, we classified network-layer applications based on the service they deliver on this layer. For example, in the M-Health, cellular components have a crucial role.
- Edge-layer: Most applications of IoT that need to interact with the appointed environment involved with edge-layer [68]. The best example of an edge-layer application is RFID.

2.2.1. Edge layer security requirements

With regards to direct interaction between IoT environments and active/passive connected things in the edge layer, we classified edge layer security requirements as authenticity, limitation in access, threat hunting and encryption of connected devices. *Authentication* Implementing a good authentication mechanism in every computational environment ensures that private data will be accessed by trusted parties. While many of the traditional algorithms and authentication mechanisms could still provide security for authenticity in the IoT environment, they stuck into trouble due to the nature of their infrastructure. Nowadays, Two-Factor Authentication has been widely adopted in many services especially in cloud-end services. Some services also use single sign-on (SSO), an authentication system that allows users to sign on to a single account, such as Google or Facebook, and uses the same token generated by the authentication mechanism to access other services. NIST has recently released a new draft of the Security and Privacy Controls, which acknowledges the benefits of combining multi-factor authentication (MFA) [32] and SSO to improve system security. With regards to privilege control, Force [69] proposed an authentication model that works in two different modes depending on the privilege of the users [70]. For basic users who only access the devices, their requests are authenticated and delegated by the IoT cloud provider. Thus, they get the processed data sent back from the cloud provider. For advanced users who can perform the firmware update, their initial authentication is performed by the cloud provider. Their proposed schemes fall short in two aspects. First, the solution does not meet the scalability concern to deal with a large number of users with multiple connected devices. Therefore, increases the latency for IoT environment clients to access their devices. Secondly, due to the resource limitation, it could be challenging for IoT devices to perform the SSO authentication and the proper timestamp validations for the advanced users. *Limited access* Limited access protection provides a limitation for the resource consumption and third-party services that edge-devices are supposed to communicate with, or access without interrupting their normal operations. One of the front-line defensive mechanisms is network access control (NAC). With NAC, the network access requests are either accepted or denied based on a pre-determined set of rules and policies that are programmed into the system. While the concept is straightforward, it is more challenging to deploy NAC because of its requirements of interacting with the protocols. Such a mechanism could be enforced in either devices or in specifically configured network devices. More often, the devices are hard-coded to connect to certain corporate domains. However, this makes them vulnerable to Domain Name Services (DNS) cache poisoning attacks. Apthorpe et al. showed that passive network spectators could infer device types and user behaviors by collecting DNS queries [71]. Therefore, DNS plays a critical role in IoT security efforts. Domain Name System Security Extensions (DNSSEC) is a suite of IETF specifications for securing the information provided by the DNS. While DNSSEC guarantees secure naming, it only has signed but not encrypted responses. In addition, DNSSEC has not been widely adopted by mainstream IoT device vendors. Even with the encryption mechanisms adopted by the devices, they are often utilized over TCP connections instead of UDP communications.

There have been many efforts aiming to raise awareness of this issue and provide resources to mitigate related threats. For example, the DNS Privacy Project [72] lists some of the current solutions, such as DNS-over-TLS (DoT), DNSover-HTTP (DoH), DNS-over-DTLS, DNSCrypt, DNS-over-HTTPS (proxied), etc. But all these protocols are designed for conventional devices with abundant resources. *Threat hunting* Besides the interactions with the external services, the IoT devices internal functionality soundness is also very crucial. Once a connected device gets compromised, all the proposed security mechanisms would fail. In the worst case, other IoT devices connected on the same network could also be infected. To mitigate such threats, both proactive and reactive mechanisms should be employed. For the proactive mechanisms, all the firmware and software updates should be examined by performing either deep packet inspection or fingerprint identifications. For the reactive mechanisms, malicious activities should trigger alert and be blocked when they happen. Both methods require cooperation and support from the network layer.

The network layer provides channels for IoT devices to interact with other entities on the Internet. At the same time, it also provides a pathway for malicious actors. Thus, the security functions in this layer should satisfy several requirements, including: (a) robust to zero-day vulnerabilities, (b) scale to a large number of connected devices, and (c) easy to manage and customize. *Encryption* To protect user privacy and the critical information (e.g., the user credential), the data exchanged between the IoT devices and the cloud provider should be properly encrypted. Conventional encryption schemes provide strong security guarantees. However, it is challenging to apply them to the resource-limited devices. To address this issue, NIST outlined a number of lightweight cryptography methods and reported that they could be useful in IoT devices [73]. In this report, they specified several performance metrics for both hardware and software. They summarized several lightweight cryptography primitives that have been proposed, including lightweight block ciphers, hash functions, Message Authentication Codes (MAC).

2.2.2. Network layer security requirements

Traffic shaping In the investigation conducted by Apthorpe et al. [71], a network traffic monitor could easily absorb the sensitive device information following three steps. First, network traffic flow could be divided into several packet streams which consist of the application or service payload by the destination or gateway IP addresses of the cloud or third-party services. Thus, traffic monitors could identify and count the distinct connected IoT devices behind a Network Address Translator (NAT). In the second step, Network monitors could identify each individual IoT device by associating DNS queries with each packet stream. Finally, simple calculations of transmit/receive rates of each stream reveals the potential user interactions with the devices.

Zhang et al. showed that IoT components are able to collect and fingerprint the wireless packets exchanged between the IoT devices and the gateway device [74]. The fingerprint of an event is defined by a cluster of packet sequences that are similar to each other. Each wireless packet in the sequence is represented by a quadruple of packet length, timestamp of the packet, source device, and destination device. Then, the similarities of the sequences are measured with Levenshtein Distance [75].

Traffic shaping is designed to be a bandwidth management technique to optimize or guarantee performance, improve latency, or increase usable bandwidth. **Traffic monitoring** Certain security functions in the edge layer rely on the information obtained from the network layer to take actions, such as malware detection. This function is mainly proposed to prevent IoT devices from being infected by malware from unknown or compromised third-party service providers. In a recent work, Alhanahnah et al. [76] proposed a method to generate signatures for IoT malware using and N-gram text analysis. In this analysis, they showed that there are specific high privilege commands and malicious DNS associated with IP address strings that are used for Command and Control (C2C) channel in the malware samples. These signatures could be organized together to produce malware detection rules.

Since malicious traffic enters the system through network standards and protocols, the security assessments are usually executed in the network layer. However, given the ruleset, it is still challenging to perform the signature matching due to applied encryption mechanisms in the edge layer. Conventionally, some systems propose to mount a man-in-the-middle attack on SSL by installing fake certificates on the middlebox [77]. However, this breaks the end-to-end security of SSL.

Anomaly detection Anomaly detection component provides another layer of protection in case the devices are infected. For detecting IoT botnet activity and its evolution since Mirai [47] Nokia released an analysis report. In this report, IoT botnets accounted for 78% of the malware carrier network activity detected in 2018. The malicious activities involve hacking activities, scanning activities, and Distributed Denial of Service (DDoS) attacks. Such attacking activities have been extensively studied and many research efforts have aimed to mitigate these attacks over the past decade. The number of connected devices makes it almost impossible to provide an effective defense mechanism close to the target. On the other hand, a security mechanism on the device side may lead to inappropriate detection due to the lack of information that exists on devices. Hence, the limited access function that implemented on the device could not be sufficient for detecting an attack behavior. We could also rely on the fact that IoT devices present certain behavior patterns for normal operations. In HoMonit [74], Zhang et al. showed that the states of the devices could be captured by the sequence of the packets. In addition, the state transitions are dictated by the automation programs installed in the service cloud. Therefore, Deterministic Finite Automation (DFA) could be used to reflect normal device behaviors. Even for devices without automation programs, such as Amazon Echo, their activity patterns should still be predictable since they are not as multi-functional as mobile devices or general-purpose machines.

2.2.3. Application layer security requirements

In the application layer of IoT architecture, based on the working environment of the domain application different functionalities are defined. However, most of these functionalities perform their tasks by APIs to interact between cloud back-end and user interface. On the other hand, most of the APIs are using a low-level security mechanism or they may not have installed a simple security mechanism at all -See Fig. 6.

Application verification As we pointed out, most of the IoT applications are supported by back-end services running on cloud platforms. They provide core services like remote administration and control, monitoring and logging, and content. Most IoT devices have the trust of the service providers by default. However, cloud endpoints are discovered to have several potential vulnerabilities, including misbehaving services, weak authentications, and insecure APIs, from previous studies. For example, Obermaier et al. analyzed four video surveillance systems and found that attackers could inject footage, trigger a false alarm, and carry out denial-of-service attacks against the camera system [78]. These threats come from weak authentications, misconfigurations of the infrastructure, and insecure APIs. The authentication could be improved by employing SSO authentications proposed in the device layer and a robust identity management system. We elaborate on the proposed security functions to address the other problems next. Fernandes et al. conducted a security analysis of the programming framework of smart homes, particularly focusing on evaluating Samsung's SmartThings platform [79]. Through their analysis, they discovered security-critical design flaws in the capability model and the event subsystem of the SmartThings cloud. The potential security violations caused by the flaws include: over-privileged access in SmartApps, insufficient sensitive event data protection, insecurity of third party integration, unsafe use of Groovy dynamic method invocation, and unrestricted API access control. Some of the vulnerabilities could be addressed with the above-mentioned mechanisms to secure APIs, while other concerns like over-privileged access and insufficient event data protection require more efforts to detect and mitigate.

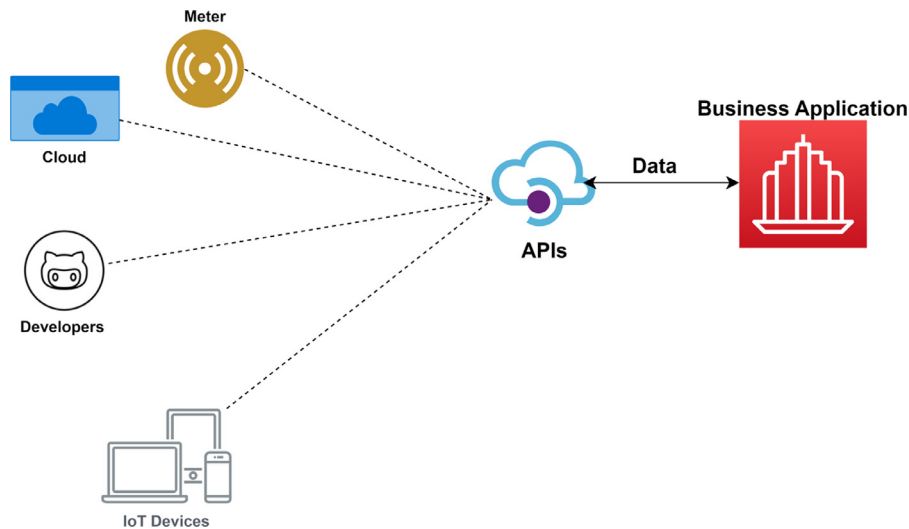


Fig. 6. IoT business application APIs functionalities .

The over-privileged accesses enable the SmartApps to gain access to all capabilities of the devices so that a malicious application could run hidden services or even take control of the whole device. The lack of protection of sensitive event data makes it possible to leak sensitive user information to other applications subscribing to the same event. Furthermore, since the integrity of the events is not protected, malicious actors could easily launch spoofing event attacks. Many research efforts have been invested to mitigate such threats. For example, Fernandes et al. proposed a framework to support flow policy rules for IoT applications [80]. Jia et al. proposed a system to gather information before a sensitive action is executed, and ask for user approval through frequent run-time prompts [81]. Some researchers also explore to address the root cause of these threats. He et al. studied the limitations of the current access control and authentication model, and they envisioned a capability-based security model for these platforms [82].

However, most existing solutions are proposed to be executed and enforced in the back-end cloud platform, which will become unreliable once the cloud gets compromised. Thus, we suggest to build a more robust monitoring system on the user end to perform integrity checks against the applications. Since the state transitions of the devices are dictated by the commands received from the applications, monitoring, and profiling the state transition patterns could be applied to achieve this goal.

Secure API APIs are the focal point of cloud innovation and enable connections and data sharing. All the applications providing services to IoT devices rely on APIs to function or grow. For example, the cloud IoT applications use APIs to gather data, or even control other devices. However, they remain the most overlooked threat in the IoT environment. This is due to the fact that API vulnerabilities are not easy to spot and require specialized technology for detection and prevention. The most common APIs in the IoT environment are Web APIs and cloud backend APIs. Simple Object Access Protocol (SOAP) and Representational State Transfer (REST) are two popular approaches for implementing APIs.

The built-in standards and envelope style of payload transport of SOAP require more overhead compared with other API implementations, such as REST. Unlike SOAP, which requires parsing and routing for each request, REST leverages standard HTTP requests. For example, Samsung SmartThings Cloud utilizes REST APIs to control and get status notifications from IoT devices. However, SOAP could provide more comprehensive security and compliance benefits.

To secure the APIs against attacks, proper authentications and authorizations are required. For authentications, OAuth2 [83] and OpenID Connect [84] are the most widely adopted mechanisms. In the device layer, users should be prevented from accessing API functions outside their predefined roles so that a read-only API client should not be allowed to access an endpoint providing administration functionality. Thus, each API call should be assigned an API token to validate incoming queries and prevent attacks on endpoints.

Information forensic Many big data analytic frameworks and models have been proposed to improve the performance and quality of the provided cloud service. Naturally, the data generated by the devices are valuable sources for security analytics. The behaviors of IoT devices are usually dictated by the automation programs, thus the state transitions should follow certain behavior model. Even though the devices are subject to the influence of the physical environment, the devices are often deployed in a static environment. Thus the collected data should present predictive patterns as well. Based on these observations, multi-dimensional security analytics that correlates data from multiple domains help service providers identify anomalies that might be suspicious, malicious, or inadvertent, and provide context intelligence regarding the nature of the threat, threat vectors used, associated business risk, and recommended mitigation.

For example, an automation application could connect smart thermometer with the smart lock on the window by enforcing a policy which is to open the window when the temperature increases above 80°F. Then, if an attacker is in the

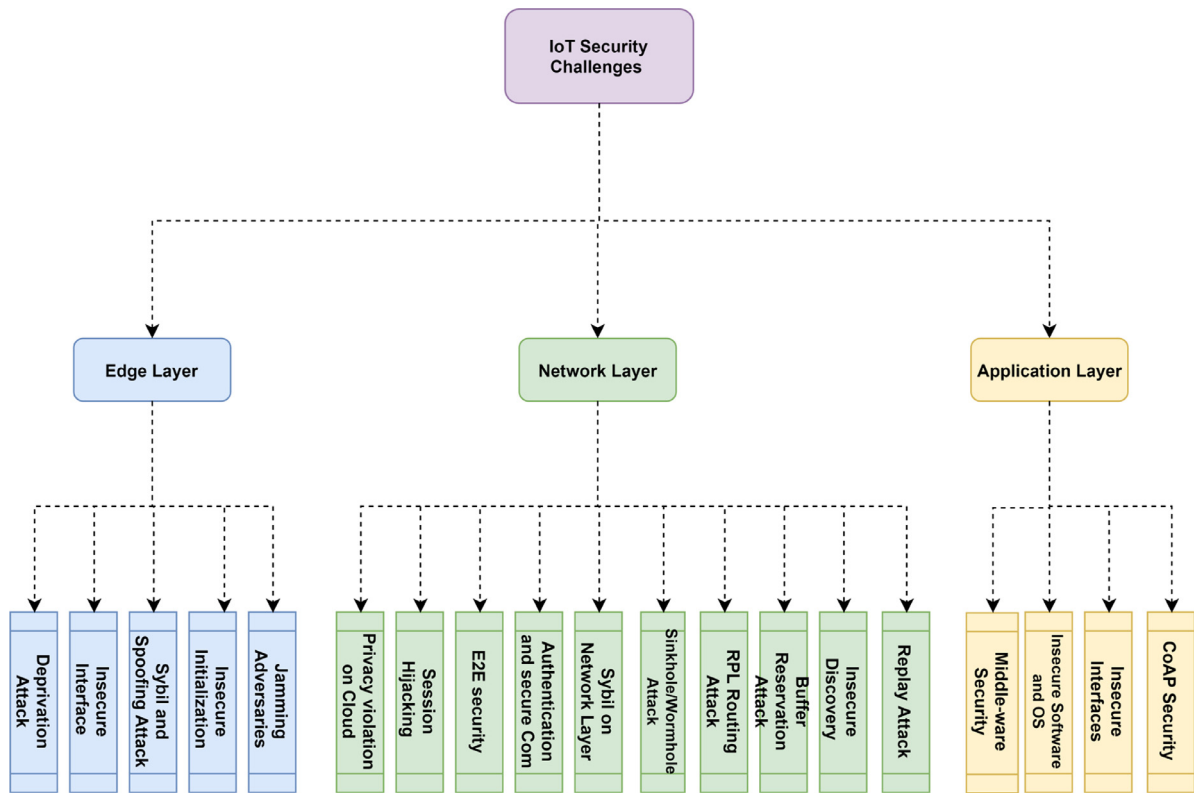


Fig. 7. IoT security challenges taxonomy based on layered architecture .

proximity of the device, it is very easy for the attacker to exploit this policy and increase the temperature of the environment. In such a situation, the cloud service should be able to monitor the state transitions of the device and associate the transitions with the behavior and pattern that learned from history as well as third party information, such as weather reports and the mobility or location of user cellphones. Another utilization of security analytics is to detect whether there has been a spike in CPU on the sensor or irregular amounts of keep-alive packets on the device. These data could indicate whether the device has exceeded its baseline of data or is performing its programmed behaviors.

3. IoT security and privacy challenges

According to the IoT properties and due to the commercialization aspect in most domain applications, IoT faces a large number of security challenges [85]. Each security challenge could be involved in one or more layers of the IoT architecture, thus each one demands significant upkeep. For example, the diversity of IoT devices in terms of their functionalities can lead to security challenges. Due to the high diversity of connected devices and their associated communication protocols in IoT, vulnerability detection is needed as a continues monitoring in the IoT environment based its defined application [86–89]. In terms of client sensitive data privacy and trust, not only the IoT environment needs a data protection mechanism to protect the IoT environment from out side intruder, but also a comprehensive mechanism is needed for inside threats. Sicari et al. [90] presented research challenges and their solution in the IoT security based on the main security concerns which were defined in eight categories. One of the challenges through implementing of the IoT environment is centralized and decentralized architecture. Roman et al. [91] conducted a research on the centralized and distributed approaches in the IoT environment. They introduced an attacker model that was applied to both centralized and distributed IoT architectures, and studied the main challenges and promising solutions in the design and deployment of the security mechanisms. Current and emerging protocols always cause security costs for their inventors. Although such protocols may implement a primitive defensive mechanism, most of their underlying mechanisms are not well-configured or switched-off. A wide range of inter-connection in IoT devices and their back-end systems creates more opportunities for adversaries to lunch their malicious files over the network.

As the IoT paradigm includes a wide range of devices and components ranging from small embedded processing chips to large high-end cloud servers, it needs to address security challenges at different layers. A taxonomy of security challenges for IoT is given in Fig. 7. As it can be seen in the figure, we classified the security challenges in the three-layered reference architecture. We also defined each existing challenge in the taxonomy, described in the following sub-sections.

3.1. Application layer security challenges

As mentioned in security requirements of the application layer, this layer is vulnerable to several attacks [16,92,93].

3.1.1. CoAP security

The Constrained Application Protocol (CoAP) operation is very similar to HTTP while using DTLS provides end-to-end security. The CoAP messages follow a predefined structure that proposed in RFC-7252 [94], using encryption systems for secure communications is essential for this protocol.

3.1.2. Insecure interfaces

Insecure interfaces are one of the common application layer security challenges. IoT service's clients generally access the desirable services via web interfaces and these interfaces are targeted as a point of vulnerability in their hosting environments [95].

3.1.3. Insecure software and OS

Software/firmware of IoT devices can be hosted the malicious application due to misconfiguration [95]. Therefore they can cause many security challenges due to weak code verifying or misconfiguration.

3.1.4. Middle-ware security

The aim of IoT middle-ware is to perform secure communication among different types of IoT components and interfaces. Therefore, different IoT environment using middlewares for interaction to other environments [17,96].

3.2. Network layer security challenges

The major concern of network layer security and privacy challenges includes secure communication, routing and session management that applied on standards and protocols of the network layer of IoT devices (e.g. transport layer). We present an attack taxonomy in the network layer which has been extracted from the most recent papers that investigate this layer. We reviewed these papers and extracted the most common attacks in recent years and ranked the as follows:

3.2.1. Replay attack

Due to the fragmentation of IPv6 packets in connected devices in the IoT environment for mapping into predefined small frame size in IEEE802.15.x standard, reconstructing fragmented packets is required in the destination device. This phenomena not only can lead to buffer overflows and device rebooting on destination side [97] but also may be transformed to Replay Attack if the malicious devices try to send duplicated fragmented packets [98].

3.2.2. Insecure nearest node discovery

To find the nearest devices in the same working network of IoT environment, Neighbor Discovery Protocol (NDP) used in IPv6. NDP finds the MAC addresses of available routers, maintains address resolution, and detects address duplication. This operation without an appropriated authentication mechanism can lead to DDoS attack [99].

3.2.3. Buffer overflow attack

In the network layer receiving nodes need to reserve an amount of space for incoming packets assembly. In some cases an intruder may exploit this mechanism by sending incomplete packets, this action can lead to DoS attack by discarding other packets because of buffer space being occupied [98].

3.2.4. RPL routing attack

The routing protocol in IPv6 of IoT environment is one vulnerable protocol for Low Power and Lossy Networks (RPL) [100]. In Sinkhole attack, a malicious device tries to show the optimum path to Base Station. By doing this, the malicious devices will be able to tamper other nodes data and also has the opportunity to modified packet payload or even drops the message which can cause delay [101].

3.2.5. Sybil attack on network layer

Fake nodes in network layer can perform Sybil attack by their fake identity in the network layer and even violate user data privacy, spamming, infiltrating malware or perform phishing attack on other nodes [102].

3.2.6. Authentication and secure communication

All IoT devices and corresponding users need to be authenticated through a key management mechanism. But sometimes computational overhead that comes from cryptographic algorithm leads to security vulnerability [16,103,104]. For example, due to resources limitation, Datagram Transport Level Security (DTLS) overhead needs to be minimized.

3.2.7. E2E security

End-to-end security provides a secure mechanism for ensuring that sender and receiver data transmission performs reliably. To reach this goal, data privacy violation will be the main security challenge for the network layer of IoT [16].

3.2.8. Session hijacking

The session hijacking can lead to denial-of-service in the IoT environment by fake messages generated by attackers [105,106]. Attacker tries to keep a session by forge messages, while the other nodes try to re-transmit the modified packet sequence number.

3.2.9. Privacy violation on the cloud

Since in most IoT environments data storage happens through back-end cloud infrastructure, IoT user data privacy such as identity or location may be violated by malicious service providers or even malicious actor on the same provider.

3.3. Edge layer security challenges

Edge layer security challenges are involved with the device level include hardware level or data link layer in the IoT environments. Security challenges of this layer can lead to system functionality failure by threat propagation into other layers.

3.3.1. Frequency jamming adversaries

Radiofrequency jamming by adversaries on a wireless node in the IoT environment can cause a huge downtime on victim environment. This security threat can be done by a radio interference affecting the network layer to render unpredictable results due to not having proper connection to a legitimate node [107–109].

3.3.2. Insecure boot-up/initialization

Since most IoT devices need to be initialized themselves after every reboot, this can be considered as a severe vulnerability if this procedure is not protected appropriately. A malicious actor can access the victim node configuration and tune it in the wrong way. Hence, the edge-layer devices need to be secure for prohibiting unauthorized access [110,111].

3.3.3. Spoofing attack

Spoofing Attack in IoT edge layer devices can happen by Sybil nodes which impersonate themselves for degrading IoT environment functionality. This attack can cause a huge denial of service for the legitimate nodes due to consuming network resources by fake nodes [59,112].

3.3.4. Insecure interfaces

The insecure interface of IoT devices can cause serious security challenges in the working environment. By insecure interfaces, all other system functionality or even defensive mechanisms in IoT devices can be compromised by a malicious node in the network or the application layer [113].

3.3.5. Deprivation attack

The deprivation attacks can be caused by a vulnerability to make an IoT edge level device keeps awake and causes energy loss for devices. This challenge can be considered as a serious one when a large number of tasks get assigned to a single node [114,115].

4. IoT security and privacy solutions

Security challenges in each layer of IoT architecture involved with all application-based (i.e. application, firmware, interfaces, etc.), network-based (i.e. cloud back-end, standards, protocols, etc.), and device-based (i.e. physical devices, cloud gateways, etc.) components of this paradigm. In this section, we present a taxonomy based on state-of-art solutions for each security challenge. Application layer solutions are presented in Table 4, Network layer solutions are presented in Table 5, and the Edge layer security solutions are presented in Table 6.

The security challenges in the application layer of IoT put the new users on a dilemma of whether applying the IoT services or not in their working environments. Fortunately, a considerable amount of research has been done to solve the common security issues in this layer. As it can be seen in Table 4, we present the proposed solutions that extracted from the recent studies for each security and privacy challenge that exists in the application layer. We also provide a propagation column for showing the impact of the challenges and corresponded solutions.

Since CoAP security issues which use clear text for messaging, most of solutions consider securing the transmission channel among users and service applications. Using DTLS and proxy mirroring techniques are the most promising solutions for handling security issues associated with CoAP. Regards to security challenges with the insecure interfaces (mostly include web interfaces), Using an encrypted channel by TLS and secure coding against Cross-Site Scripting (XSS) and SQL injection

Table 4

The presented security solutions for the application layer of IoT environment.

Challenge	Affected layer(s)	Solution	application	Reference
CoAP security with internet	Application layer, and network layer	Using DTLS, secure application Proxy and Resource Directory	[20]	[16,19,93,116]
Insecure interfaces	Application layer	Password strength verification, secure coding (SQLi and XSS), and installing application gateway firewalls	[20]	[54,114,117]
Insecure Firmware/OS	Application layer and network layer	Regular secure updates of software/firmware, use of file signatures, and encryption with validation	[54]	[118,119]
Middleware security	Application layer, transport layer, and network layer	Secure communication channel with authentication, define security policies, key management and distribution, install secure gateways & M2M components, lightweight encryption systems	[120]	[120–122]

Table 5

The presented security solutions for the network layer of IoT environment.

Challenge	Affected layer(s)	Solution	application	Reference
Replay Attack	Network layer	define timestamp and authentication parameter for packets verification, define checksum by hash value	[123]	[98,124–126]
Insecure nearest node discovery	Network layer	Authentication by encrypted (ECC) based signatures	[127]	[128,129]
Buffer Overflow attack	Network layer	installing threat hunting modules like IDS	[130]	[131,132]
RPL routing attack	Network layer	authentication via lightweight encryption system, and monitoring connected devices	[133]	[100,134]
Sinkhole and Wormhole attacks	Network layer	verification using hash systems, trust level management, device communication analysis, anomaly detection via IDS, using encrypted key management, signal strength monitoring	[135]	[136–138]
Sybil attacks	Network layer	Graph analysis, user interaction analysis, Applying access control list	[102]	[102,139–141]
Authentication and secure communication	Network layer	light weight ticket granting system, applying symmetric and asymmetric encryption system for encrypting packet payload dispatch type values with, collecting logs	[142]	[136,137,140,143–145]
end-to-end security	Network layer	installing IPSec, applying advance encryption system for authentication and authorization	[146]	[147,148]
Session Hijacking	Network layer	using secret key for long-time session, light-weight encryption system	[149]	[16,70,105]

Table 6

The presented security solutions for the edge layer of IoT environment.

Challenge	Affected layer(s)	Solution	application	Reference
Frequency Jamming adversaries	Edge-Layer	Measuring signal strength, computing packet delivery ratio, encoding packets with error correcting codes, and change of frequencies and locations	[150,151]	[107–109]
Spoofing Attack	Edge-Layer	Signal strength measurements, and channel estimation	[152]	[110,112]
Insecure initialization and configuration	Edge-Layer	Setting data transmission rates b/w nodes, and introducing artificial noise	[153]	[107–109]
Insecure interface	Edge-Layer	hardware based TPM modules, and avoiding testing/debugging tools	[47]	[51,113]
Sleep deprivation attack	Edge-Layer and network Layer	Multi-layer based intrusion detection system	[154]	[124,155,156]

vulnerabilities are the most promising solutions. As we mentioned earliler due to the low computational ability of IoT devices, the firmware and Operating System (OS) of these devices operate under the lightweight mechanism and this issue causes security challenges. Using a lightweight encryption system and update the firmware frequently can lead to provide a safe performance by working device. The security challenges among IoT devices (the connected things) is another common issue in most IoT domain applications particularly transportation and mobility environments. Authentication via secure communication which offers key management, secure gateway, secure policies is the key solution for middleware security challenges in the application layer.

In conclusion, the common security issues in the application layer can be handled by two important factors as follows: a) providing secure communication and b) using secure coding technique and application firewall along encryption systems for most of the operational procedures in the application layer of IoT environments.

As mentioned in security and privacy challenges and as it can be seen from Table 5, the network layer of IoT faces a wide range of cyber threats. Since IoT devices need to operate and communicate without human intervention, most of the security challenges are related to IoT nodes communications vulnerabilities.

Despite Replay Attack is one of the common security challenges in the IoT network devices, but by defining time stamp and verification parameters like checksum by hash value are good actions to prevent this threat. Using a lightweight version of encryption systems can significantly decrease most of the serious threats in the network layer. For example, RPL routing attack, Session Hijacking, secure authentication, and preserving privacy by end-to-end security are the high-risk challenges that can be avoided by encryption systems and checksum. Installing defensive mechanism system like network firewall and Intrusion Detection System in the network layer can always decrease the impact of security challenges. For instance, Sinkhole and wormhole attack and end-to-end security can be mitigated or even eradicated by installing an appropriated IDS. In recent years, Applying AI techniques in IDS has led to higher detection rates and lower false alarms respectively [138].

In summary, there three key solutions for the network layer security and privacy challenges in the IoT environments: (a) Implementing appropriate (lightweight) encryption systems to maintain conventionality and integrity of the data based on the domain application of the IoT (b) Installing good tracing parameters like timestamp for the authorizing the transferred packets (c) Implementing an appropriate threat hunting module for detecting threats and mitigating destructive propagation.

Most of the edge-layer security challenges in IoT environments happen due to lack of direct access to manage and maintain the connected devices. Moreover, dispersion of deployment of the edge-layer devices in each IoT application environment causes critical security issues -See Table 6. However, we can mitigate the risk of the security challenges by applying an appropriate mechanism. For example, to overcome the issues that cause service disruption, like Frequency Jamming and Spoofing attack, signal measurement and packet coding along with channel estimation for ensuring trusted parties for further communication are the promising solutions. Although edge-layer devices have limited computational resources, deploying AI agents [157] for detecting any anomalies, especially in coordinator device, can mitigate security risk like Sleep Deprivation attack.

Overall, although most of the security issues in the edge-layer may lead to denial of service in the IoT environments, some key solutions can mitigate this critical issue including (a) signal measurement for detecting adversarial attempts (b) threat hunting module by applying the AI agent for detecting any misbehavior by connected devices.

5. Conclusion

With increasing the usage of IoT devices in most business domains and personal lives, security concerns get higher. Due to the limitation in resources and diversity of components in different IoT environments, a wide range of vulnerabilities have emerged. Most of these vulnerabilities can lead to system failure in the working environment of the IoT. Since there is no predefined standard for IoT environment, most of the conducted researches to date have presented IoT security challenges and solutions without any specific structure. In this paper, we conducted a comprehensive survey on IoT security requirements, challenges, and solutions based on the security properties of the common IoT environment in a three-layer architectural structure. We also provided a taxonomy for existing IoT security requirements and challenges in Application, Network, and Edge layers and presented the corresponding solutions for those challenges. The main aim of this survey was to provide an IoT cyber threat taxonomy for fellow IoT practitioners and researchers to conduct their future works based on each challenge and solution.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] J. Lin, W. Yu, N. Zhang, X. Yang, H. Zhang, W. Zhao, A survey on internet of things: architecture, enabling technologies, security and privacy, and applications, *IEEE Internet Things J.* 4 (5) (2017) 1125–1142, doi:[10.1109/JIOT.2017.2683200](https://doi.org/10.1109/JIOT.2017.2683200).
- [2] M. Chiang, T. Zhang, Fog and IoT: an overview of research opportunities, *IEEE Internet Things J.* 3 (6) (2016) 854–864.
- [3] P. Ray, A survey on internet of things architectures, *J. King Saud Univ.* 30 (3) (2018) 291–319, doi:[10.1016/j.jksuci.2016.10.003](https://doi.org/10.1016/j.jksuci.2016.10.003).
- [4] X. Jia, Q. Feng, T. Fan, Q. Lei, RFID technology and its applications in internet of things (IoT), in: *2012 2nd International Conference on Consumer Electronics, Communications and Networks (CECNet)*, IEEE, 2012, pp. 1282–1285.
- [5] N. Binti A Kamaludeen, S.P. Lee, R. M. Parizi, Guideline-based approach for IoT home application development, in: *2019 International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData)*, 2019, pp. 929–936, doi:[10.1109/iThings/GreenCom/CPSCom/SmartData.2019.00165](https://doi.org/10.1109/iThings/GreenCom/CPSCom/SmartData.2019.00165).
- [6] M. Yun, B. Yuxin, Research on the architecture and key technology of internet of things (IoT) applied on smart grid, in: *2010 International Conference on Advances in Energy Engineering*, IEEE, 2010, pp. 69–72.
- [7] J. Sakhnini, H. Karimipour, A. Dehghantanha, R.M. Parizi, G. Srivastava, Security aspects of internet of things aided smart grids: a bibliometric survey, *Internet Things* (2019) 100111, doi:[10.1016/j.iot.2019.100111](https://doi.org/10.1016/j.iot.2019.100111).
- [8] T.M. Behera, S.K. Mohapatra, U.C. Samal, M.S. Khan, M. Daneshmand, A.H. Gandomi, Residual energy-based cluster-head selection in WSNS for IoT application, *IEEE Internet Things J.* 6 (3) (2019) 5132–5139, doi:[10.1109/JIOT.2019.2897119](https://doi.org/10.1109/JIOT.2019.2897119).
- [9] L. Catarinucci, D. De Donno, L. Mainetti, L. Palano, L. Patrono, M.L. Stefanizzi, L. Tarricone, An IoT-aware architecture for smart healthcare systems, *IEEE Internet Things J.* 2 (6) (2015) 515–526.

- [10] G. Srivastava, R.M. Parizi, A. Dehghantanha, K.-K.R. Choo, Data sharing and privacy for patient IoT devices using blockchain, in: The 7th International Conference on Smart City and Informatization (iSCI 2019), 2019, pp. 1–15, doi:10.1007/978-981-15-1301-5_27.
- [11] W. He, G. Yan, L. Da Xu, Developing vehicular data cloud services in the IoT environment, *IEEE Trans. Ind. Inf.* 10 (2) (2014) 1587–1595.
- [12] A. Paranjothi, U. Tanik, Y. Wang, M.S. Khan, Hybrid-vehfog: a robust approach for reliable dissemination of critical messages in connected vehicles, *Trans. Emerg. Telecommun. Technol.* 30 (6) (2019) e3595, doi:10.1002/ett.3595. E3595 ETT-18-0175.R3
- [13] A. Paranjothi, M.S. Khan, S. Zeadally, A. Pawar, D. Hicks, Gstr: secure multi-hop message dissemination in connected vehicles using social trust model, *Internet Things* 7 (2019) 100071, doi:10.1016/j.iot.2019.100071.
- [14] A.H. Ngu, M. Gutierrez, V. Metsis, S. Nepal, Q.Z. Sheng, IoT middleware: a survey on issues and enabling technologies, *IEEE Internet Things J.* 4 (1) (2016) 1–20.
- [15] J. Qi, P. Yang, G. Min, O. Amft, F. Dong, L. Xu, Advanced internet of things for personalised healthcare systems: a survey, *Pervasive Mob. Comput.* 41 (2017) 132–149.
- [16] J. Granjal, E. Monteiro, J.S. Silva, Application-layer security for the WoT: extending CoAP to support end-to-end message security for internet-integrated sensing applications, in: International Conference on Wired/Wireless Internet Communication, Springer, 2013, pp. 140–153.
- [17] V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, J. Alonso-Zarate, A survey on application layer protocols for the internet of things, *Trans. IoT Cloud Comput.* 3 (1) (2015) 11–17.
- [18] J. Choi, Y. In, C. Park, S. Seok, H. Seo, H. Kim, Secure IoT framework and 2D architecture for End-To-End security, *J. Supercomput.* 74 (8) (2018) 3521–3535.
- [19] S. Arvind, V.A. Narayanan, An overview of security in CoAP: attack and analysis, in: 2019 5th International Conference on Advanced Computing & Communication Systems (ICACCS), IEEE, 2019, pp. 655–660.
- [20] A. Bhattacharjya, X. Zhong, J. Wang, X. Li, CoAP—application layer connection-less lightweight protocol for the internet of things (IoT) and CoAP-IPSEC security with DTLS supporting CoAP, in: Digital Twin Technologies and Smart Cities, Springer, 2020, pp. 151–175.
- [21] J. Santos, J.J. Rodrigues, B.M. Silva, J. Casal, K. Saleem, V. Denisov, An IoT-based mobile gateway for intelligent personal assistants on mobile health environments, *J. Netw. Comput. Appl.* 71 (2016) 194–204.
- [22] B. Ali, A. Awad, Cyber and physical security vulnerability assessment for IoT-based smart homes, *Sensors* 18 (3) (2018) 817.
- [23] M.A. Khan, K. Salah, IoT security: review, blockchain solutions, and open challenges, *Fut. Gener. Comput. Syst.* 82 (2018) 395–411, doi:10.1016/j.future.2017.11.022.
- [24] D. Burke, Preventing DDOS Attacks Against IoT Devices, Utica College, 2018 PhD Thesis.
- [25] D. Minoli, B. Occhiogrosso, Blockchain mechanisms for IoT security, *Internet Things* 1 (2018) 1–13.
- [26] A. Molina Zarca, J. Bernal Bernabe, I. Farris, Y. Khettab, T. Taleb, A. Skarmeta, Enhancing IoT security through network softwarization and virtual security appliances, *Int. J. Netw. Manage.* 28 (5) (2018) e2038.
- [27] J. Chen, C. Touati, Q. Zhu, Optimal secure two-layer IoT network design, *IEEE Trans. Control Netw. Syst.* (2019).
- [28] P. Mahalle, S. Babar, N.R. Prasad, R. Prasad, Identity management framework towards internet of things (IoT): roadmap and key challenges, in: International Conference on Network Security and Applications, Springer, 2010, pp. 430–439.
- [29] D. Puthal, S. Nepal, R. Ranjan, J. Chen, Threats to networking cloud and edge datacenters in the Internet of Things, *IEEE Cloud Comput.* 3 (3) (2016) 64–71.
- [30] J. Canedo, A. Skjellum, Using machine learning to secure IoT systems, in: 2016 14th Annual Conference on Privacy, Security and Trust (PST), IEEE, 2016, pp. 219–222.
- [31] E.M. Dovom, A. Azmoodeh, A. Dehghantanha, D.E. Newton, R.M. Parizi, H. Karimipour, Fuzzy pattern tree for edge malware detection and categorization in IoT, *J. Syst. Archit.* 97 (2019) 1–7, doi:10.1016/j.sysarc.2019.01.017.
- [32] P. Grassi, M. Garcia, J. Fenton, DRAFT NIST Special Publication 800-63-3 Digital Identity Guidelines, National Institute of Standards and Technology, Los Altos, CA, 2017.
- [33] J. Ren, H. Guo, C. Xu, Y. Zhang, Serving at the edge: a scalable IoT architecture based on transparent computing, *IEEE Netw.* 31 (5) (2017) 96–105.
- [34] B. Ur, J. Jung, S. Schechter, The current state of access control for smart devices in homes, Workshop on Home Usable Privacy and Security (HUPS), HUPS 2014, 2013.
- [35] J. Wurm, K. Hoang, O. Arias, A.-R. Sadeghi, Y. Jin, Security analysis on consumer and industrial IoT devices, in: 2016 21st Asia and South Pacific Design Automation Conference (ASP-DAC), IEEE, 2016, pp. 519–524.
- [36] K. Sha, R. Errabelly, W. Wei, T.A. Yang, Z. Wang, Edgesec: design of an edge layer security service to enhance IoT security, in: 2017 IEEE 1st International Conference on Fog and Edge Computing (ICFEC), IEEE, 2017, pp. 81–88.
- [37] D. Costa, I. Silva, L. Guedes, F. Vasques, P. Portugal, Availability issues in wireless visual sensor networks, *Sensors* 14 (2) (2014) 2795–2821.
- [38] W. Trappe, R. Howard, R.S. Moore, Low-energy security: limits and opportunities in the internet of things, *IEEE Secur. Privacy* 13 (1) (2015) 14–21.
- [39] K. Georgiou, S. Xavier-de-Souza, K. Eder, The IoT energy challenge: a software perspective, *IEEE Embed. Syst. Lett.* 10 (3) (2017) 53–56.
- [40] N. Vidgren, K. Haataja, J.L. Patino-Andres, J.J. Ramirez-Sanchis, P. Toivanen, Security threats in ZigBee-enabled systems: vulnerability evaluation, practical experiments, countermeasures, and lessons learned, in: 2013 46th Hawaii International Conference on System Sciences, IEEE, 2013, pp. 5132–5138.
- [41] Morgner P, Mattejat S, Benenson Z. All your bulbs are belong to us: Investigating the current state of security in connected lighting systems, arXiv preprint arXiv:1608.03732 2016 Aug 12.
- [42] G. Ho, D. Leung, P. Mishra, A. Hosseini, D. Song, D. Wagner, Smart locks: lessons for securing commodity internet of things devices, in: Proceedings of the 11th ACM on Asia Conference on Computer and Communications Security, ACM, 2016, pp. 461–472.
- [43] R. Roman, C. Alcaraz, J. Lopez, N. Sklavos, Key management systems for sensor networks in the context of the internet of things, *Comput. Electric. Eng.* 37 (2) (2011) 147–159.
- [44] N.E. Petroulakis, E.Z. Tragou, A.G. Fragkiadakis, G. Spanoudakis, A lightweight framework for secure life-logging in smart environments, *Inf. Secur. Tech. Rep.* 17 (3) (2013) 58–70.
- [45] M.A. Simplicio Jr, M.V. Silva, R.C. Alves, T.K. Shibata, Lightweight and escrow-less authenticated key agreement for the internet of things, *Comput. Commun.* 98 (2017) 43–51.
- [46] B. Ghena, W. Beyer, A. Hillaker, J. Pevanek, J.A. Halderman, Green lights forever: analyzing the security of traffic infrastructure, 8th USENIX Security Workshop on Offensive Technologies (WOOTS'14), 2014.
- [47] C. Kolias, G. Kambourakis, A. Stavrou, J. Voas, DDoS in the IoT: mirai and other botnets, *Computer* 50 (7) (2017) 80–84.
- [48] C. Schuett, J. Butts, S. Dunlap, An evaluation of modification attacks on programmable logic controllers, *Int. J. Crit. Infrastruct. Prot.* 7 (1) (2014) 61–68.
- [49] D.E. Kouicem, A. Bouabdallah, H. Lakhlef, Internet of things security: a top-down survey, *Comput. Netw.* 141 (2018) 199–221, doi:10.1016/j.comnet.2018.03.012.
- [50] C. Konstantinou, M. Maniatakis, Impact of firmware modification attacks on power systems field devices, in: 2015 IEEE International Conference on Smart Grid Communications (SmartGridComm), IEEE, 2015, pp. 283–288.
- [51] F.A. Alaba, M. Othman, I.A.T. Hashem, F. Alotaibi, Internet of things security: a survey, *J. Netw. Comput. Appl.* 88 (2017) 10–28, doi:10.1016/j.jnca.2017.04.002.
- [52] K. Zhao, L. Ge, A survey on the internet of things security, in: 2013 Ninth International Conference on Computational Intelligence and Security, IEEE, 2013, pp. 663–667.
- [53] J. Guo, I.-R. Chen, J.J. Tsai, A survey of trust computation models for service management in internet of things systems, *Comput. Commun.* 97 (2017) 1–14, doi:10.1016/j.comcom.2016.10.012.

- [54] N. Neshenko, E. Bou-Harb, J. Crichigno, G. Kaddoum, N. Ghani, Demystifying IoT security: an exhaustive survey on IoT vulnerabilities and a first empirical look on internet-scale IoT exploitations, *IEEE Commun. Surv. Tut.* (2019).
- [55] C. Sarkar, S.A.U. Nambi, R.V. Prasad, A. Rahim, A scalable distributed architecture towards unifying IoT applications, in: 2014 IEEE World Forum on Internet of Things (WF-IoT), IEEE, 2014, pp. 508–513.
- [56] C.-L. Zhong, Z. Zhu, R.-G. Huang, Study on the IOT architecture and gateway technology, in: 2015 14th International Symposium on Distributed Computing and Applications for Business Engineering and Science (DCABES), IEEE, 2015, pp. 196–199.
- [57] J. Lloret, J. Tomas, A. Canovas, L. Parra, An integrated IoT architecture for smart metering, *IEEE Commun. Mag.* 54 (12) (2016) 50–57.
- [58] S.A. Al-Qaseemi, H.A. Almulhim, M.F. Almulhim, S.R. Chaudhry, IoT architecture challenges and issues: lack of standardization, in: 2016 Future Technologies Conference (FTC), IEEE, 2016, pp. 731–738.
- [59] B. Cheng, D. Zhu, S. Zhao, J. Chen, Situation-aware IoT service coordination using the event-driven SOA paradigm, *IEEE Trans. Netw. Serv. Manage.* 13 (2) (2016) 349–361, doi:10.1109/TNSM.2016.2541171.
- [60] L. Wang, L. Da Xu, Z. Bi, Y. Xu, Data cleaning for RFID and WSN integration, *IEEE Trans. Ind. Inf.* 10 (1) (2013) 408–418.
- [61] T.M. Behera, S.K. Mohapatra, U.C. Samal, M.S. Khan, Hybrid heterogeneous routing scheme for improved network performance in WSNS for animal tracking, *Internet Things* 6 (2019) 100047, doi:10.1016/j.iot.2019.03.001.
- [62] T.M. Behera, S.K. Mohapatra, U.C. Samal, M.S. Khan, M. Daneshmand, A.H. Gandomi, I-sep: an improved routing protocol for heterogeneous WSN for IoT based environmental monitoring, *IEEE Internet Things J.* (2019), doi:10.1109/JIOT.2019.2940988. 1–1
- [63] A.A. Patel, S.J. Soni, A novel proposal for defending against vampire attack in WSN, in: 2015 Fifth International Conference on Communication Systems and Network Technologies, IEEE, 2015, pp. 624–627.
- [64] A. Whitmore, A. Agarwal, L. Da Xu, The internet of things—a survey of topics and trends, *Inf. Syst. Front.* 17 (2) (2015) 261–274, doi:10.1007/s10796-014-9489-2.
- [65] M. Abdel-Basset, G. Manogaran, M. Mohamed, Internet of things (IoT) and its impact on supply chain: a framework for building smart, secure and efficient systems, *Fut. Gener. Comput. Syst.* 86 (2018) 614–628, doi:10.1016/j.future.2018.04.051.
- [66] M.M. Rathore, A. Paul, W.-H. Hong, H. Seo, I. Awan, S. Saeed, Exploiting IoT and big data analytics: defining smart digital city using real-time urban data, *Sustain. Cities Soc.* 40 (2018) 600–610, doi:10.1016/j.scs.2017.12.022.
- [67] L. Atzori, R. Girau, V. Pilloni, M. Uras, Assignment of sensing tasks to IoT devices: exploitation of a social network of objects, *IEEE Internet Things J.* 6 (2) (2019) 2679–2692, doi:10.1109/JIOT.2018.2873501.
- [68] A. Kumari, S. Tanwar, S. Tyagi, N. Kumar, R.M. Parizi, K.-K.R. Choo, Fog data analytics: a taxonomy and process model, *J. Netw. Comput. Appl.* 128 (2019) 90–104, doi:10.1016/j.jnca.2018.12.013.
- [69] J.T. Force, Security and Privacy Controls for Information Systems and Organizations, Technical Report, National Institute of Standards and Technology, 2017.
- [70] L. Barreto, A. Celesti, M. Villari, M. Fazio, A. Puliafito, An authentication model for IoT clouds, in: 2015 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM), IEEE, 2015, pp. 1032–1035.
- [71] A. Apthorpe, N. Reisman, D. Feamster, N. A smart home is no castle: Privacy vulnerabilities of encrypted IoT traffic. *arXiv preprint arXiv:1705.06805*. 2017 May 18.
- [72] DNS Privacy Project Homepage - DNS Privacy Project - Global Site.
- [73] K. McKay, L. Bassham, M. Sönmez Turan, N. Mouha, Report on Lightweight Cryptography, Technical Report, National Institute of Standards and Technology, 2016.
- [74] W. Zhang, Y. Meng, Y. Liu, X. Zhang, Y. Zhang, H. Zhu, Homonit: monitoring smart home apps from encrypted traffic, in: Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security, ACM, 2018, pp. 1074–1088.
- [75] L. Yujian, L. Bo, A normalized levenshtein distance metric, *IEEE Trans. Pattern Anal. Mach. Intell.* 29 (6) (2007) 1091–1095.
- [76] M. Alhanahnah, Q. Lin, Q. Yan, N. Zhang, Z. Chen, Efficient signature generation for classifying cross-architecture IoT malware, in: 2018 IEEE Conference on Communications and Network Security (CNS), IEEE, 2018, pp. 1–9.
- [77] L.S. Huang, A. Rice, E. Ellingsen, C. Jackson, Analyzing forged SSL certificates in the wild, in: 2014 IEEE Symposium on Security and Privacy, IEEE, 2014, pp. 83–97.
- [78] J. Obermaier, M. Huttle, Analyzing the security and privacy of cloud-based video surveillance systems, in: Proceedings of the 2nd ACM International Workshop on IoT Privacy, Trust, and Security, ACM, 2016, pp. 22–28.
- [79] E. Fernandes, J. Jung, A. Prakash, Security analysis of emerging smart home applications, in: 2016 IEEE Symposium on Security and Privacy (SP), IEEE, 2016, pp. 636–654.
- [80] E. Fernandes, J. Paupore, A. Rahmati, D. Simionato, M. Conti, A. Prakash, Flowfence: practical data protection for emerging IoT application frameworks, in: 25th USENIX Security Symposium (USENIX Security 16), 2016, pp. 531–548.
- [81] Y.J. Jia, Q.A. Chen, S. Wang, A. Rahmati, E. Fernandes, Z.M. Mao, A. Prakash, S.J. University, ContextIoT: towards providing contextual integrity to applied IoT platforms., *NDSS*, 2017.
- [82] W. He, M. Golla, R. Padhi, J. Ofek, M. Dürmuth, E. Fernandes, B. Ur, Rethinking access control and authentication for the home internet of things (IoT), in: 27th USENIX Security Symposium (USENIX Security 18), 2018, pp. 255–272.
- [83] D. Hardt, The OAuth 2.0 Authorization Framework, 2012.
- [84] D. Recordon, D. Reed, OpenID 2.0: a platform for user-centric identity management, in: Proceedings of the Second ACM Workshop on Digital Identity Management, ACM, 2006, pp. 11–16.
- [85] A. Alrawais, A. Alhothaily, C. Hu, X. Cheng, Fog computing for the internet of things: security and privacy issues, *IEEE Internet Comput.* 21 (2) (2017) 34–42, doi:10.1109/MIC.2017.37.
- [86] V. Sivaraman, H.H. Gharakheili, A. Vishwanath, R. Boreli, O. Mehani, Network-level security and privacy control for smart-home IoT devices, in: 2015 IEEE 11th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob), IEEE, 2015, pp. 163–167.
- [87] F. Olivier, G. Carlos, N. Florent, New security architecture for IoT network, *Procedia Comput. Sci.* 52 (2015) 1028–1033.
- [88] T. Baranwal, P.K. Pateriya, Development of IoT based smart security and monitoring devices for agriculture, in: 2016 6th International Conference—Cloud System and Big Data Engineering (Confluence), IEEE, 2016, pp. 597–602.
- [89] W.-C. Chien, C.-F. Lai, H.-H. Cho, H.-C. Chao, A SDN-SFC-based service-oriented load balancing for the IoT applications, *J. Netw. Comput. Appl.* 114 (2018) 88–97, doi:10.1016/j.jnca.2018.04.009.
- [90] S. Sicari, A. Rizzardi, L.A. Grieco, A. Coen-Porisini, Security, privacy and trust in internet of things: the road ahead, *Comput. Netw.* 76 (2015) 146–164.
- [91] R. Roman, J. Zhou, J. Lopez, On the features and challenges of security and privacy in distributed internet of things, *Comput. Netw.* 57 (10) (2013) 2266–2279.
- [92] M. Brachmann, S.L. Keoh, O.G. Morchon, S.S. Kumar, End-to-end transport security in the IP-based internet of things, in: 2012 21st International Conference on Computer Communications and Networks (ICCCN), IEEE, 2012, pp. 1–5.
- [93] M. Sethi, J. Arkko, A. Keränen, End-to-end security for sleepy smart object networks, in: 37th Annual IEEE Conference on Local Computer Networks—Workshops, IEEE, 2012, pp. 964–972.
- [94] Z. Shelby, K. Hartke, C. Bormann, The Constrained Application Protocol (Coap)(RfC 7252), 2014. Jun-2014 Available online. <http://www.rfc-editor.org/info/rfc7252>
- [95] M. Zolanvari, R. Jain, IoT Security: A Survey, 2015.
- [96] D. Conzon, T. Bolognesi, P. Brizzi, A. Lotito, R. Tomasi, M.A. Spirito, The virtus middleware: an Xmp based architecture for secure IoT communications, in: 2012 21st International Conference on Computer Communications and Networks (ICCCN), IEEE, 2012, pp. 1–6.

- [97] H. Kim, Protection against packet fragmentation attacks at 6lowpan adaptation layer, in: 2008 International Conference on Convergence and Hybrid Information Technology, IEEE, 2008, pp. 796–801.
- [98] R. Hummen, J. Hiller, H. Wirtz, M. Henze, H. Shafagh, K. Wehrle, 6LoWPAN fragmentation attacks and mitigation mechanisms, in: Proceedings of the Sixth ACM Conference on Security and Privacy in Wireless and Mobile Networks, ACM, 2013, pp. 55–66.
- [99] B. Park, Threats and security analysis for enhanced secure neighbor discovery protocol (SEND) of IPv6 NDP security, *Int. J. Control Autom.* 4 (4) (2011).
- [100] A. Dvir, T. Holczer, L. Buttyan, VeRA - version number and rank authentication in RPL, in: 2011 IEEE Eighth International Conference on Mobile Ad-Hoc and Sensor Systems, 2011, pp. 709–714, doi:10.1109/MASS.2011.76.
- [101] C. Pu, S. Hajjar, Mitigating forwarding misbehaviors in RPL-based low power and lossy networks, in: 2018 15th IEEE Annual Consumer Communications Networking Conference (CCNC), 2018, pp. 1–6, doi:10.1109/CCNC.2018.8319164.
- [102] K. Zhang, X. Liang, R. Lu, X. Shen, Sybil attacks and their defenses in the internet of things, *IEEE Internet Things J.* 1 (5) (2014) 372–383.
- [103] S. Raza, S. Duquenois, J. Höglund, U. Roedig, T. Voigt, Secure communication for the internet of things—a comparison of link-layer security and IPsec for 6LoWPAN, *Secur. Commun. Netw.* 7 (12) (2014) 2654–2668.
- [104] J. Granjal, E. Monteiro, J.S. Silva, Network-layer security for the internet of things using TinyOS and BLIP, *Int. J. Commun. Syst.* 27 (10) (2014) 1938–1963.
- [105] M.H. Ibrahim, Octopus: an edge-fog mutual authentication scheme., *IJ Netw. Secur.* 18 (6) (2016) 1089–1101.
- [106] N. Park, N. Kang, Mutual authentication scheme in secure internet of things technology for comfortable lifestyle, *Sensors* 16 (1) (2016) 20.
- [107] Y. Li, L. Shi, P. Cheng, J. Chen, D.E. Quevedo, Jamming attacks on remote state estimation in cyber-physical systems: a game-theoretic approach, *IEEE Trans. Automat. Control* 60 (10) (2015) 2831–2836.
- [108] S. Vadhvani, B. Eksioğlu, H. Meda, A. Nandi, Jamming attacks on wireless networks: a taxonomic survey, *Int. J. Prod. Econ.* 172 (2016) 76–94.
- [109] Y. Guan, X. Ge, Distributed attack detection and secure estimation of networked cyber-physical systems against false data injection attacks and jamming attacks, *IEEE Trans. Signal Inf. Process. Netw.* 4 (1) (2017) 48–59.
- [110] Y.-W.P. Hong, P.-C. Lan, C.-C.J. Kuo, Enhancing physical-layer secrecy in multiantenna wireless systems: an overview of signal processing approaches, *IEEE Signal Process. Mag.* 30 (5) (2013) 29–40.
- [111] S.H. Chae, W. Choi, J.H. Lee, T.Q. Quek, Enhanced secrecy in stochastic wireless networks: artificial noise with secrecy protected zone, *IEEE Trans. Inf. Forensics Secur.* 9 (10) (2014) 1617–1628.
- [112] L. Xiao, L.J. Greenstein, N.B. Mandayam, W. Trappe, Channel-based detection of sybil attacks in wireless networks, *IEEE Trans. Inf. Forensics Secur.* 4 (3) (2009) 492–503, doi:10.1109/TIFS.2009.2026454.
- [113] I. Andrea, C. Chrysostomou, G. Hadjichristofi, Internet of things: security vulnerabilities and challenges, in: 2015 IEEE Symposium on Computers and Communication (ISCC), IEEE, 2015, pp. 180–187.
- [114] H. Xie, Z. Yan, Z. Yao, M. Atiquzzaman, Data collection for security measurement in wireless sensor networks: a survey, *IEEE Internet Things J.* 6 (2) (2018) 2205–2224.
- [115] O.A. Osanaiye, A.S. Alfa, G.P. Hancke, Denial of service defence for resource availability in wireless sensor networks, *IEEE Access* 6 (2018) 6975–7004.
- [116] R.H. Randhawa, A. Hameed, A.N. Mian, Energy efficient cross-layer approach for object security of CoAP for IoT devices, *Ad Hoc Netw.* 92 (2019) 101761.
- [117] D. Wang, J. Ming, T. Chen, X. Zhang, C. Wang, Cracking IoT device user account via brute-force attack to SMS authentication code, in: Proceedings of the First Workshop on Radical and Experiential Security, ACM, 2018, pp. 57–60.
- [118] A. Mohanty, I. Obaidat, F. Yilmaz, M. Sridhar, Control-hijacking vulnerabilities in IoT firmware: a brief survey, in: Proceedings of the 1st International Workshop on Security and Privacy for the Internet-of-Things (IoTSec), 2018.
- [119] W. Zhou, Y. Jia, A. Peng, Y. Zhang, P. Liu, The effect of IoT new features on security and privacy: new threats, existing solutions, and challenges yet to be solved, *IEEE Internet Things J.* 6 (2) (2018) 1606–1616.
- [120] R.V. Reddy, D. Murali, J. Rajeshwar, Context-aware middleware architecture for IoT-based smart healthcare applications, in: Innovations in Computer Science and Engineering, Springer, 2019, pp. 557–567.
- [121] C.H. Liu, B. Yang, T. Liu, Efficient naming, addressing and profile services in internet-of-things sensory environments, *Ad Hoc Netw.* 18 (2014) 85–101.
- [122] V. Hassija, V. Chamola, V. Saxena, D. Jain, P. Goyal, B. Sikdar, A survey on IoT security: application areas, security threats, and solution architectures, *IEEE Access* 7 (2019) 82721–82743.
- [123] S. Kalra, S.K. Sood, Secure authentication scheme for IoT and cloud servers, *Pervasive Mob. Comput.* 24 (2015) 210–223.
- [124] O. Brun, Y. Yin, E. Gelenbe, Deep learning with dense random neural network for detecting attacks against IoT-connected home environments, *Procedia Comput. Sci.* 134 (2018) 458–463.
- [125] P. Gope, R. Amin, S.H. Islam, N. Kumar, V.K. Bhalla, Lightweight and privacy-preserving RFID authentication scheme for distributed IoT infrastructure with secure localization services for smart city environment, *Fut. Gener. Comput. Syst.* 83 (2018) 629–637.
- [126] J. Pacheco, S. Hariri, Anomaly behavior analysis for IoT sensors, *Trans. Emerg. Telecommun. Technol.* 29 (4) (2018) e3188.
- [127] O. Novo, Blockchain meets IoT: an architecture for scalable access management in IoT, *IEEE Internet Things J.* 5 (2) (2018) 1184–1195.
- [128] M. Malik, M. Dutta, Defending DDoS in the insecure internet of things: a survey, in: Artificial Intelligence and Evolutionary Computations in Engineering Systems, Springer, 2018, pp. 223–233.
- [129] R. Doshi, N. Apthorpe, N. Feamster, Machine learning DDoS detection for consumer internet of things devices, in: 2018 IEEE Security and Privacy Workshops (SPW), IEEE, 2018, pp. 29–35.
- [130] Y. Lee, J. Jeong, Y. Son, Design and implementation of the secure compiler and virtual machine for developing secure IoT services, *Fut. Gener. Comput. Syst.* 76 (2017) 350–357, doi:10.1016/j.future.2016.03.014.
- [131] F.A. Teixeira, F.M. Pereira, H.-C. Wong, J.M. Nogueira, L.B. Oliveira, SIoT: Securing Internet of Things through distributed systems analysis, *Fut. Gener. Comput. Syst.* 92 (2019) 1172–1186.
- [132] W. Han, X. Liu, H. Zhang, R. Quan, L. Shen, Dynamically-enabled defense effectiveness evaluation of IoT based on vulnerability analysis, in: Proceedings of the 3rd International Conference on Multimedia Systems and Signal Processing, ACM, 2018, pp. 99–103.
- [133] D. Airehour, J. Gutierrez, S.K. Ray, Secure routing for internet of things: a survey, *J. Netw. Comput. Appl.* 66 (2016) 198–213.
- [134] A. Le, J. Loo, A. Lasebae, A. Vinel, Y. Chen, M. Chai, The impact of rank attack on network topology of routing protocol for low-power and lossy networks, *IEEE Sens. J.* 13 (10) (2013) 3685–3692.
- [135] N. Tsitsiroudi, P. Sarigiannidis, E. Karapistoli, A.A. Economides, EyeSim: a mobile application for visual-assisted wormhole attack detection in IoT-enabled WSNs, in: 2016 9th IFIP Wireless and Mobile Networking Conference (WMNC), IEEE, 2016, pp. 103–109.
- [136] K. Weekly, K. Pister, Evaluating sinkhole defense techniques in RPL networks, in: 2012 20th IEEE International Conference on Network Protocols (ICNP), IEEE, 2012, pp. 1–6.
- [137] F. Ahmed, Y.-B. Ko, Mitigation of black hole attacks in routing protocol for low power and lossy networks, *Secur. Commun. Netw.* 9 (18) (2016) 5143–5154.
- [138] H.H. Pajouh, R. Javidan, R. Khayami, D. Ali, K.-K.R. Choo, A two-layer dimension reduction and two-tier classification model for anomaly-based intrusion detection in IoT backbone networks, *IEEE Trans. Emerg. Top. Comput.* (2016).
- [139] S. Sharmila, G. Umamaheswari, Detection of sinkhole attack in wireless sensor networks using message digest algorithms, in: 2011 International Conference on Process Automation, Control and Computing, IEEE, 2011, pp. 1–6.
- [140] Cao Q, Yang X. SybilFence: Improving social-graph-based sybil defenses with user negative feedback. *arXiv preprint arXiv:1304.3819*. 2013 Apr 13.
- [141] N. Maheshwari, H. Dagale, Secure communication and firewall architecture for IoT applications, in: 2018 10th International Conference on Communication Systems Networks (COMSNETS), 2018, pp. 328–335, doi:10.1109/COMSNETS.2018.8328215.

- [142] K.T. Nguyen, M. Laurent, N. Oualha, Survey on secure communication protocols for the Internet of Things, *Ad Hoc Netw.* 32 (2015) 17–31, doi:[10.1016/j.adhoc.2015.01.006](https://doi.org/10.1016/j.adhoc.2015.01.006).
- [143] L. Alvisi, A. Clement, A. Epasto, S. Lattanzi, A. Panconesi, Sok: the evolution of sybil defense via social networks, in: 2013 IEEE Symposium on Security and Privacy, IEEE, 2013, pp. 382–396.
- [144] A. Mohaisen, N. Hopper, Y. Kim, Keep your friends close: incorporating trust into social network-based sybil defenses, in: 2011 Proceedings IEEE INFOCOM, IEEE, 2011, pp. 1943–1951.
- [145] M. Wazid, A.K. Das, S. Kumari, M.K. Khan, Design of sinkhole node detection mechanism for hierarchical wireless sensor networks, *Secur. Commun. Netw.* 9 (17) (2016) 4596–4614.
- [146] S.R. Moosavi, T.N. Gia, E. Nigussie, A.M. Rahmani, S. Virtanen, H. Tenhunen, J. Isoaho, End-to-end security scheme for mobility enabled healthcare Internet of Things, *Fut. Gener. Comput. Syst.* 64 (2016) 108–124, doi:[10.1016/j.future.2016.02.020](https://doi.org/10.1016/j.future.2016.02.020).
- [147] S. Raza, T. Chung, S. Duquennoy, T. Voigt, U. Roedig, Securing Internet of Things with Lightweight IPsec, 2010.
- [148] S. Raza, S. Duquennoy, J. Höglund, U. Roedig, T. Voigt, Secure communication for the internet of things—a comparison of link-layer security and IPsec for 6LoWPAN, *Secur. Commun. Netw.* 7 (12) (2014) 2654–2668.
- [149] Q. Hu, B. Du, K. Markantonakis, G.P. Hancke, A session hijacking attack against a device-assisted physical layer key agreement, *IEEE Trans. Ind. Inf.* (2019), doi:[10.1109/TII.2019.2923662](https://doi.org/10.1109/TII.2019.2923662). 1–1
- [150] S. Sciancalepore, G. Oliveri, R. Di Pietro, Strength of crowd (SOC)—defeating a reactive jammer in IoT with decoy messages, *Sensors* 18 (10) (2018) 3492.
- [151] X. Tang, P. Ren, Z. Han, Jamming mitigation via hierarchical security game for IoT communications, *IEEE Access* 6 (2018) 5766–5779.
- [152] L. Wallgren, S. Raza, T. Voigt, Routing attacks and countermeasures in the RPL-based internet of things, *Int. J. Distrib. Sens. Netw.* 9 (8) (2013) 794326.
- [153] Y. Liu, J. Briones, R. Zhou, N. Magotra, Study of secure boot with a FPGA-based IoT device, in: 2017 IEEE 60th International Midwest Symposium on Circuits and Systems (MWSCAS), IEEE, 2017, pp. 1053–1056.
- [154] T. Bhattasali, R. Chaki, S. Sanyal, Sleep deprivation attack detection in wireless sensor network, *IJCA* 40 (15) (2012) 19–25, doi:[10.5120/5056-7374](https://doi.org/10.5120/5056-7374).
- [155] J. Sherry, C. Lan, R.A. Popa, S. Ratnasamy, Blindbox: deep packet inspection over encrypted traffic, *ACM SIGCOMM Comput. Commun. Rev.* 45 (4) (2015) 213–226.
- [156] O. Brun, Y. Yin, J. Augusto-Gonzalez, M. Ramos, E. Gelenbe, *IoT Attack Detection with Deep Learning*, 2018.
- [157] H. HaddadPajouh, A. Dehghantanha, R. Khayami, K.-K.R. Choo, A deep Recurrent Neural Network based approach for Internet of Things malware threat hunting, *Fut. Gener. Comput. Syst.* 85 (2018) 88–96.